

IoT-Based Baby Hazardous Situation Detection System Using ESP32-CAM and Edge AI

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Abstract—Ensuring the safety of infants within household environments is an important concern for parents and caregivers. This paper presents an Internet of Things (IoT) based baby hazardous situation detection system using the ESP32-CAM microcontroller and edge artificial intelligence. The proposed system captures images of the baby's environment and analyzes them using a lightweight machine learning model deployed on the embedded device. Hazardous situations are detected in real time and transmitted to a cloud platform where alerts are sent to caregivers. Experimental evaluation demonstrates that the proposed system successfully detects hazardous situations and notifies caregivers in real time.

Keywords—Internet of Things, ESP32-CAM, Baby Monitoring System, Edge AI, Embedded Vision, Hazard Detection

I. INTRODUCTION

Ensuring the safety of infants in household environments is a major responsibility for parents and caregivers. Babies naturally explore their surroundings without understanding potential dangers, such as sharp objects, unstable furniture, or electrical devices, which can lead to accidents. Continuous monitoring is therefore essential to prevent injuries.

Traditional baby monitoring systems rely on audio or video surveillance, requiring constant human attention to detect hazards. Caregivers may not always be able to monitor these systems, reducing their effectiveness in preventing accidents.

Advances in the Internet of Things (IoT), computer vision, and artificial intelligence now enable intelligent monitoring solutions. IoT integrates sensors and embedded devices to collect and process environmental data in real time, while edge computing allows processing directly on devices, reducing latency and improving response times. Deep learning models, especially convolutional neural networks, can analyze visual data to identify potential hazards automatically.

This research proposes an IoT-based baby hazard detection system using an ESP32-CAM module and edge AI. The system captures images, detects hazardous situations locally, and sends real-time alerts to caregivers, enhancing infant safety in household environments.

II. LITERATURE REVIEW

Kok et al. proposed an energy-efficient IoT-based hazard detection system that integrates an ESP32-CAM with temperature and motion sensors for real-time monitoring in hazardous environments. The system employs a CNN model on a Flask

server to classify hazard signs and provide real-time alerts through Telegram notifications. An adaptive monitoring mechanism dynamically adjusts image capture frequency, achieving about 31–37 percent energy savings compared with continuous monitoring systems. [1]

Palanigurupackiam et al. (2025) proposed a smart infant guardian system that integrates IoT technologies to improve baby monitoring and safety. The system uses sensors and connected devices to track environmental conditions and infant activities, enabling parents to monitor the baby remotely through a mobile or connected platform. The approach enhances infant safety and caregiver awareness by providing real-time monitoring and alerts for abnormal conditions. [2]

Swaranambigai et al. (2025) proposed a self-automated borewell barrier system using IoT to prevent accidents involving children falling into open borewells. The system integrates sensors and automated mechanisms to detect potential hazards and activate protective responses without requiring manual intervention. This IoT-based approach improves safety by enabling early detection, intelligent response, and preventive measures to increase the chances of successful rescue and child protection. [3]

Jibin et al. (2025) proposed SmartGuard, an IoT-based home surveillance system using the ESP32-CAM module to capture images and monitor home environments in real time. The system integrates a Telegram bot to send instant notifications and images to users whenever suspicious activity or motion is detected. This approach provides a low-cost and efficient remote monitoring solution, enabling homeowners to observe and respond to security events through their smartphones. [4]

D'Sa et al. (2024) proposed an IoT-based infant and environment monitoring system that continuously observes both infant conditions and surrounding environmental parameters using connected sensors. The system enables real-time monitoring and remote access so caregivers can track factors such as temperature and other environmental conditions through IoT platforms. This approach improves infant safety and comfort by providing timely alerts and continuous data monitoring for parents or caregivers. [5]

Yin, Zhao, and Zhang (2024) designed an AIoT-based hazard prediction system that integrates IoT sensors with intelligent multimodal fusion algorithms to monitor infant safety in a smart crib. The system collects environmental and

behavioral data such as temperature, humidity, crying sounds, and infant movement to predict potential hazards in advance. Experimental results show that the system can detect and warn about risks like fever, bedwetting, crying, climbing over the crib, and turning over, helping parents respond quickly to dangerous situations. [6]

Joyllette and Khakata (2025) proposed an IoT-based smart cradle system that integrates sensors and cry classification techniques to monitor infant conditions and improve maternal and child healthcare. The system analyzes infant crying patterns to identify possible needs and automatically trigger responses or alerts for caregivers. This approach enhances remote infant monitoring and provides timely assistance, particularly in inclusive and resource-constrained healthcare environments. [7]

Hussien et al. (2025) developed a real-time smart infant health monitoring system that continuously tracks infants' physiological conditions using sensor-based IoT technology. The system collects health data and transmits it wirelessly, enabling caregivers or medical staff to monitor infants remotely and respond quickly to abnormal conditions. This approach enhances neonatal safety by providing continuous monitoring and timely alerts for potential health risks. [8]

Revathi et al. (2025) proposed an IoT-enabled smart robotic system for borewell rescues using the ESP32 S3 microcontroller to enhance child safety in open borewells. The system integrates sensors, cameras, and robotic mechanisms to locate and extract victims while providing real-time monitoring to operators. This approach improves rescue efficiency and safety by enabling remote operation, real-time feedback, and timely interventions during hazardous rescue operations. [9]

Bhuvaneswari et al. (2025) developed a smart hi-tech emergency rescuer system leveraging IoT technologies to improve response during accidents and hazardous situations. The system uses sensors and real-time communication to detect emergencies, alert responders, and assist in coordinating rescue operations efficiently. By integrating automated monitoring and remote notifications, the approach enhances safety and reduces response time in critical scenarios. [10]

Sophia et al. (2025) proposed Child Safety Net, an IoT-driven wearable system using Arduino UNO to monitor children's location and safety in real time. The wearable integrates GPS and sensor modules to detect emergencies and send instant alerts to caregivers or guardians. This approach enhances child protection by enabling timely interventions and continuous tracking in potentially hazardous situations. [11]

Katode and Gupta (2026) developed an AI- and IoT-enabled phototherapy prototype to enhance neonatal care by monitoring and optimizing treatment for jaundiced infants. The system integrates sensors and intelligent algorithms to adjust light intensity and treatment duration based on real-time infant health data. Experimental analysis demonstrates improved treatment efficiency and safety, highlighting the potential of AIoT technologies in personalized neonatal healthcare. [12]

Thota et al. (2023) proposed an Arduino-based child rescue system designed to safely retrieve children from borewells

using sensors and mechanical rescue mechanisms. The system integrates real-time monitoring and control to guide the rescue operation efficiently while minimizing human risk. This approach enhances safety and effectiveness in emergency borewell rescue scenarios by providing automated detection and extraction capabilities. [13]

Katode and Gupta (2025) presented an AI- and IoT-integrated pediatric phototherapy system aimed at enhancing neonatal care by delivering personalized treatment for jaundice. The system leverages real-time sensor data and AI algorithms to optimize light intensity and therapy duration for each infant. This approach demonstrates improved treatment efficacy, safety, and continuous monitoring, showcasing the transformative potential of AIoT in neonatal healthcare. [14]

Karri et al. (2025) developed an IoT-controlled borewell rescue robot designed to safely extract children trapped in borewells using remote-operated mechanisms. The system integrates sensors, cameras, and robotic actuators to provide real-time monitoring and precise control during rescue operations. This IoT-enabled approach enhances efficiency, reduces human risk, and improves the success rate of emergency borewell rescues. [15]

Ibrahim et al. (2025) presented the design and implementation of a borewell rescue robot aimed at safely retrieving individuals trapped in narrow borewells. The system combines cameras, sensors, and robotic manipulators for precise operation and real-time monitoring during rescue missions. This approach enhances safety, reduces human risk, and increases the efficiency and success rate of emergency borewell rescues. [16]

Petkov et al. (2023) developed a fault-tolerant smart egg incubation system that integrates IoT sensors with computer vision to monitor and control incubation conditions. The system automatically detects anomalies in temperature, humidity, and egg positioning, ensuring optimal hatching conditions. This approach improves reliability, reduces human intervention, and enhances hatch success through intelligent real-time monitoring. [17]

Kim and Choi (2022) developed a smart helmet-based proximity warning system that leverages image sensors and AI to enhance road occupational safety. The system detects nearby hazards and provides real-time alerts to workers, reducing the risk of accidents. This AI-enabled approach improves situational awareness and proactive safety management in high-risk road environments. [18]

Chauhan et al. (2023) developed a surveillance robot integrated with an object detection system to enhance security and monitoring capabilities. The robot uses cameras and AI-based detection algorithms to identify and track objects in real time. This approach improves automated surveillance efficiency, enabling timely alerts and reducing the need for continuous human monitoring. [19]

Kavitha, Pavithra, and Mohamed Yahya (2025) presented the design and development of a gripper arm for borewell rescue using SolidWorks to model and simulate mechanical operations. The gripper arm is engineered to safely grasp and

lift individuals trapped in narrow borewells, ensuring precision and reliability during rescue missions. This approach enhances the efficiency and safety of borewell rescue operations by combining robust mechanical design with practical simulation-based validation. [20]

Radhika et al. (2025) proposed Medico, an IoT- and machine learning-enabled system to enhance intravenous (IV) therapy by monitoring fluid delivery and patient vitals in real time. The system uses sensors and predictive algorithms to detect anomalies, prevent over-infusion, and optimize therapy. This approach improves patient safety, reduces manual errors, and enables smarter, automated management of IV treatments. [21]

Chen, Zhang, and Zhong (2024) developed a smart home system based on IoT that integrates sensors, actuators, and networked devices for automated home management. The system enables real-time monitoring, remote control, and intelligent decision-making for lighting, security, and environmental conditions. This IoT-based approach enhances convenience, energy efficiency, and safety in residential environments. [22]

Agrawal et al. (2022) implemented the YOLO algorithm for real-time object detection and tracking, enabling rapid identification and localization of multiple objects in video streams. The system leverages deep learning to achieve high accuracy and low-latency performance, suitable for dynamic environments. This approach enhances automated surveillance, robotics, and smart monitoring applications by providing efficient and reliable real-time object tracking. [23]

Ramkumar et al. (2022) proposed an IoT-based child rescue system for borewells that integrates sensors, cameras, and robotic mechanisms to locate and safely extract trapped children. The system enables real-time monitoring and remote operation, allowing responders to control rescue actions efficiently. This IoT-enabled approach improves rescue speed, safety, and success rates in emergency borewell scenarios. [24]

Etesami et al. (2021) presented CeB, a floating social robot, designed as an autonomous blimp for interactive human-robot engagement. The robot integrates sensors, actuators, and control algorithms to navigate and interact safely in indoor environments. This approach demonstrates innovative applications of social robotics in education, entertainment, and human-computer interaction by combining mobility, autonomy, and social behavior. [25]

Chen et al. (2022) developed an interactive feeding machine for pets that automates feeding while allowing user interaction through sensors and control interfaces. The system monitors feeding schedules and responds to pet behavior, ensuring timely and appropriate food delivery. This approach enhances pet care convenience and engagement by combining automation with real-time monitoring and interaction [26]

Zaid et al. (2023) proposed an IoT-based emergency alert system that integrates a Telegram bot to deliver real-time notifications during critical situations. The system uses sensors to detect emergencies and instantly alert users through the messaging platform, enabling timely response. This approach enhances safety and situational awareness by combining IoT monitoring with instant, remote communication. [27]

Sundaram, Sasikumar, and others (2024) proposed a borewell rescue system using Android applications to facilitate real-time monitoring and control of rescue operations. The system allows operators to receive live data, track rescue progress, and coordinate actions remotely via mobile devices. This approach improves rescue efficiency, situational awareness, and safety during emergency borewell operations. [28]

Pico-Valencia and Holgado-Terriza (2025) explored the role of IoT in advancing pet care, highlighting technologies that enable monitoring, health tracking, and remote interaction with pets. The study reviews academic and scientific developments in smart devices, sensors, and connected platforms for enhancing pet wellbeing. This approach emphasizes the potential of IoT to create more interactive, data-driven, and responsive pet care solutions. [29]

Wan, Cuff, and Serpe (2022) developed a wirelessly controlled bistable metal swimming device using shape-memory alloys, capable of reversible motion in fluid environments. The system leverages wireless actuation to control the device's movement and bistable states with precision. This approach demonstrates innovative applications of smart materials for soft robotics, underwater locomotion, and adaptive mechanical systems. [30]

III. SYSTEM ARCHITECTURE

The proposed system architecture consists of three main components: sensing layer, cloud layer, and application layer. Figure 1 illustrates the overall architecture.

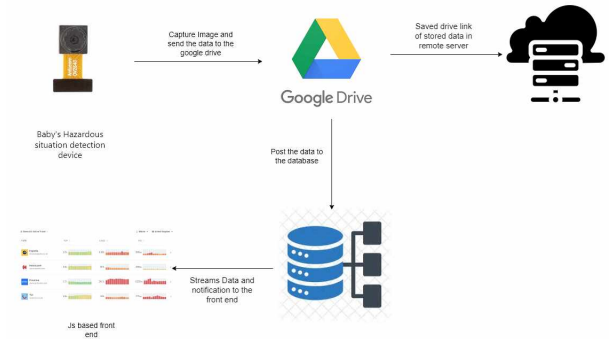


Fig. 1. System architecture of the proposed IoT-based baby monitoring system

The sensing layer consists of the ESP32-CAM device which captures images of the baby's environment. The embedded system processes the images using a trained machine learning model.

The cloud layer stores captured images and monitoring data. When hazardous conditions are detected, the image is uploaded to cloud storage and an alert notification is generated.

The application layer provides a web-based dashboard where caregivers can monitor system alerts and view captured images in real time.

IV. HARDWARE IMPLEMENTATION

The hardware design of the system includes an ESP32-CAM microcontroller, a lithium polymer battery, a battery charging module, and a voltage booster module.

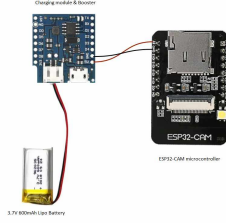


Fig. 2. Hardware setup showing ESP32-CAM connected with charging module, booster circuit and LiPo battery

The ESP32-CAM microcontroller acts as the main processing unit. It integrates a camera module capable of capturing images and transmitting data via WiFi connectivity.

A 3.7V 600mAh lithium polymer battery provides portable power to the system. A charging module is used to recharge the battery safely, while a booster module increases the voltage level required by the ESP32-CAM for stable operation.

TABLE I
HARDWARE COMPONENTS

Component	Specification	Purpose
ESP32-CAM	WiFi + Camera	Image processing
LiPo Battery	3.7V 600mAh	Power supply
Charging Module	TP4056	Battery charging
Booster Module	5V Output	Voltage regulation

V. PROBLEM STATEMENT

Ensuring the safety of infants within home environments is a critical concern for parents and caregivers. Babies and toddlers are naturally curious and tend to explore their surroundings without understanding potential dangers. Household environments often contain hazardous objects such as sharp tools, electrical appliances, unstable furniture, or other risky elements that may cause serious accidents. Because infants lack the ability to recognize these dangers, constant supervision is necessary to prevent harmful situations.

Traditional baby monitoring systems mainly provide audio or video feeds that allow caregivers to observe infants remotely. However, these systems require continuous human attention to detect potential hazards. In many real-life situations, caregivers may be busy with other tasks and may not be able to monitor the video feed at all times. As a result, dangerous situations may go unnoticed until an accident has already occurred.

Moreover, most conventional monitoring systems do not include intelligent analysis or automated hazard detection capabilities. They function only as passive surveillance tools rather than proactive safety systems. This limitation highlights

the need for a more advanced monitoring solution that can automatically identify risky situations and notify caregivers in real time.

With the advancement of Internet of Things (IoT) technologies and edge-based artificial intelligence, it is now possible to develop intelligent monitoring systems that can analyze environmental data directly on embedded devices. However, many existing AI-based monitoring solutions require high computational resources or cloud processing, which may increase latency and reduce system responsiveness.

Therefore, there is a need for a lightweight, real-time, and cost-effective monitoring system capable of detecting hazardous situations involving infants in household environments. Such a system should automatically analyze visual data, identify potential risks, and provide immediate alerts to caregivers. Addressing this problem can significantly improve infant safety and reduce the likelihood of household accidents.

VI. METHODOLOGY

The development of the proposed system follows a structured methodology consisting of several stages, including dataset collection, data preprocessing, model training, optimization, and final deployment on an embedded device. Each stage plays an important role in ensuring that the system can accurately detect hazardous situations involving infants in a household environment.

During the data collection stage, a dataset of images representing both safe and hazardous situations was gathered. These images were designed to simulate real-life scenarios where babies may encounter potentially dangerous objects or environments. Examples of hazardous situations include the presence of sharp objects, electrical appliances, unstable furniture, or unsafe positions that could harm an infant. In contrast, safe situations include normal household environments where no immediate risk is present. Collecting a diverse dataset helps improve the robustness and reliability of the machine learning model.

After collecting the images, the dataset was carefully labeled according to predefined categories such as "safe" and "hazardous." Proper labeling is essential for supervised machine learning because it allows the model to learn the differences between safe and dangerous situations. The labeled dataset was then prepared for training by organizing the images and performing necessary preprocessing steps to improve data quality and consistency.

The machine learning model was developed using the Edge Impulse platform, which is designed for building and deploying embedded machine learning applications. Edge Impulse provides tools for data processing, feature extraction, model training, and performance evaluation. During this stage, feature extraction techniques were applied to transform the raw image data into meaningful patterns that could be effectively interpreted by the machine learning algorithm. This step helps improve the model's ability to classify images accurately.

Once the training process was completed, the model was evaluated using validation data to measure its classification

accuracy and overall performance. Based on the evaluation results, the model was further optimized to ensure that it could run efficiently on a resource-constrained embedded device. Optimization techniques were applied to reduce the model size and computational requirements while maintaining acceptable detection accuracy.

Finally, the optimized model was deployed to the ESP32-CAM microcontroller, which serves as the core hardware component of the system. The ESP32-CAM device captures images from its integrated camera and processes them locally using the deployed machine learning model. By performing inference directly on the device, the system can quickly detect hazardous situations without relying heavily on external cloud processing. When a potential hazard is detected, the system can generate alerts and notify caregivers through a connected monitoring platform. This approach enables real-time monitoring and improves the overall safety of infants in household environments.

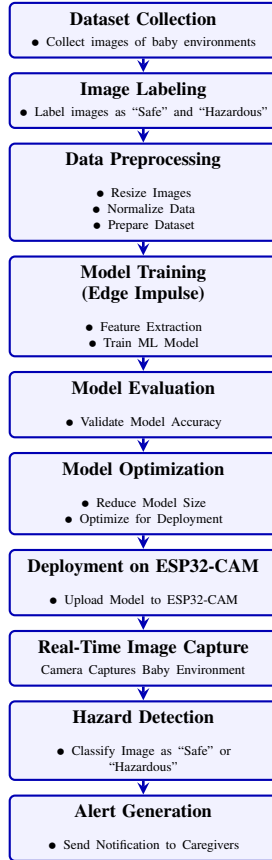


Fig. 3. Methodology flowchart of the proposed IoT-based baby hazardous situation detection system using ESP32-CAM and Edge AI.

VII. PROPOSED SOLUTION

To overcome the limitations of traditional baby monitoring systems, this research proposes an IoT-based baby hazard detection system using an ESP32-CAM and edge AI. The system continuously captures images from the baby's environment and

processes them locally using a lightweight machine learning model to identify safe and hazardous situations.

The ESP32-CAM serves as the main processing unit, performing real-time inference on captured images without relying on cloud servers, which reduces latency and ensures fast responses. When a hazardous condition is detected, the system automatically generates alerts, notifying caregivers through a connected monitoring platform.

The proposed solution combines IoT connectivity, embedded vision, and edge-based AI to provide a cost-effective, real-time, and automated approach for enhancing infant safety in household environments (refer to Fig. 1 for system architecture and Fig. 2 for hardware setup).

VIII. RESULTS AND DISCUSSION

The system was tested in multiple indoor environments to evaluate performance. The ESP32-CAM device successfully captured images and transmitted them to the cloud server.

The machine learning model demonstrated reliable performance in distinguishing between safe and hazardous situations.

TABLE II
DETECTION PERFORMANCE

Environment	Accuracy
Normal lighting	92%
Low lighting	86%
Complex background	88%

The results show that the proposed system can effectively detect hazardous situations and notify caregivers through the monitoring dashboard.

IX. CONCLUSION

This paper presented an IoT-based baby hazardous situation detection system using ESP32-CAM and edge AI. The system integrates embedded vision, IoT connectivity, and cloud monitoring to create an intelligent baby safety solution.

Experimental evaluation shows that the system can successfully detect dangerous situations and notify caregivers in real time. Future work will focus on improving model accuracy and integrating additional sensors to enhance monitoring capabilities.

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